



NORTHERN WOLVES
AIR CONDITIONING

Ref. Quote: 3112 / Date: 10/1/2026 / Rev 0

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Project: **Sunrise Rye Brook - Independent Living / Assisted Living**

Address: **900 King St, Rye Brook, NY 10573**

Mechanical drawings: **GMP set dated 07.15.2026 / Base: Quote 3112 (drawings 06.05.2026)**

Value Engineering Log - Summary

VE	Item	Deduct / Add	Electric	Approvals needed
VE-1	VAV reheat: hot water coils to electric reheat	\$23,500	+185 kW (EC)	IMEG redesign / energy code; EE and utility service capacity
VE-2	Cabinet heaters: hot water to electric	-\$18,000	+70 kW (EC)	IMEG; EE
VE-3	Basement FCUs: delete, heat / cool only where needed	-\$45,800	+30 kW (EC)	IMEG; fire protection designer for freeze-risk rooms
VE-4	Garage ramp snow melt: delete hydronic snow melt	-\$41,600	0 / +240 kW if electric by EC	Owner decision; civil / landscape for drainage
VE-5	Pool and spa: dedicated gas heaters in lieu of HX-1 / HX-2	\$15,100	0	Pool designer; NY All-Electric Buildings Act check for new gas appliances
VE-6	Delete boiler plant and hot water / glycol distribution	-\$1,091,500	-	IMEG; Sunrise
VE-9	Fire/smoke dampers at apartment duct branches	-\$299,900	-	IMEG / HKS code analysis (AL Group I-1, smoke barriers), building department
VE-14	Apartment bath exhaust ducted to DOAS exhaust	-\$13,300	-11 kW	IMEG; DOAS exhaust airflow check
VE-11	No BMS: standalone equipment controls only (Sunrise Northport precedent)	TBD		Deduct = BMS amount carried in quote 3112 less standalone controls. Needs the BMS contractor's breakdown. Supersedes VE-7.
VE-12	Common areas: delete 47 VAV boxes; single-zone RTUs for large spaces, ductless / VRF for small rooms	TBD		Replaces VE-1 if taken. Priced after IMEG issues a zoning concept.
VE-10	Shorten DOAS corridor return	TBD		Included in VE-14 (return trunk reused as exhaust).
	Recommended package: VE-1, VE-2, VE-3, VE-4, VE-5, VE-6, VE-9, VE-14	-\$1,471,500		plus VE-11 / VE-12 when priced

Notes

- Budget pricing from quantities on the GMP mechanical set dated 07.15.2026. Piping on the basement sheets is not drawn; those runs are allowances. Final pricing after IMEG issues revised drawings.
- Negative amounts are deducts from quote 3112, positive amounts are adds. VE-6 is available only together with VE-1 to VE-5. VE-12 replaces VE-1 if taken.
- Electrical work for electric heat (power, disconnects, service upgrade) by EC and is not included. Electrical, fire alarm and GC credits (damper power, TX-A circuits, chimney enclosures, combustion air areaaway) are additional to the amounts shown.
- Gas piping excluded (as in quote 3112). Redesign, re-filing and permit fees by others.

- VE-7 (packaged controls) is superseded by VE-11. VE-8 (full VRF for common areas) not recommended.

Value Engineering Log - Detail

VE-1 VAV reheat: hot water coils to electric reheat

Drawings: M2014-M2016, M6001 VAV schedule

47 VAV boxes with hot water reheat (VAV-01 to VAV-47, 629 MBH). Replace hot water coils, coil piping packages and control valve installation with factory electric reheat coils with SCR control.

Line item	Qty	Unit	Material	MH	Labor	Services
Hot water reheat coils at VAV boxes	-47	ea	-\$21,150			
Coil piping packages (valves, balancing valve, strainer, unions)	-47	ea	-\$16,450	-188	-\$11,280	
Install BMS-furnished control valves	-47	ea		-94	-\$5,640	
Electric reheat coils with SCR control, factory mounted	47	ea	\$63,450	141	\$8,460	
Subtotal			\$25,850	-141	-\$8,460	
Overhead 20%, misc 2%, tax 8.875% on material						\$6,110
VE-1 total (rounded)						\$23,500

Electric: +185 kW (EC). Approvals: IMEG redesign / energy code; EE and utility service capacity.

VE-2 Cabinet heaters: hot water to electric

Drawings: M2014-M2016, M6001 cabinet heater schedule

CH-1 to CH-12 at entries and stairs (234 MBH). Replace with electric cabinet heaters with integral thermostats, no BMS points.

Line item	Qty	Unit	Material	MH	Labor	Services
Hot water cabinet heaters CH-1 to CH-12 (CH-9 large)	-12	ea	-\$27,360	-96	-\$5,760	
Hot water piping hook-ups at cabinet heaters	-12	ea	-\$2,160	-48	-\$2,880	
Electric cabinet heaters with integral thermostat	12	ea	\$19,800	72	\$4,320	
Subtotal			-\$9,720	-72	-\$4,320	
Overhead 20%, misc 2%, tax 8.875% on material						-\$3,960
VE-2 total (rounded)						-\$18,000

Electric: +70 kW (EC). Approvals: IMEG; EE.

VE-3 Basement FCUs: delete, heat / cool only where needed

Drawings: M2000, M2002, M5000, M6000 FCU schedule

FCU-A (20, hot water/glycol heating only) and RAD-1/2 in basement back-of-house rooms. DOAS branch ducts connect directly to diffusers. Electric heaters only for freeze protection; ductless AC and thermostatic exhaust for electrical rooms, MDF and laundry (not on the drawings today).

Line item	Qty	Unit	Material	MH	Labor	Services
FCU-A fan coil units, horizontal, hot water	-20	ea	-\$56,000	-200	-\$12,000	
FCU piping hook-ups (valves, unions, flex)	-20	ea	-\$4,400	-120	-\$7,200	
RAD-1, RAD-2	-2	ea	-\$4,000	-16	-\$960	
Electric unit / wall heaters with integral thermostat	8	ea	\$7,200	40	\$2,400	
Ductless heat pump AC, MDF / electrical rooms, with line set	4	ea	\$26,000	96	\$5,760	
Thermostatic exhaust fans, laundry / electrical, with duct and louver	3	ea	\$5,400	36	\$2,160	
Subtotal			-\$25,800	-164	-\$9,840	
Overhead 20%, misc 2%, tax 8.875% on material						-\$10,160
VE-3 total (rounded)						-\$45,800

Electric: +30 kW (EC). Approvals: IMEG; fire protection designer for freeze-risk rooms.

VE-4 Garage ramp snow melt: delete hydronic snow melt

Drawings: M5000 (HX-3, P-3/P-4 120 F glycol to north and south ramps)

Delete glycol mains and manifolds to the ramps. Owner options: no snow melt (trench drains, salting) or electric snow melt cable by EC. In-slab tubing not included in this deduct.

Line item	Qty	Unit	Material	MH	Labor	Services
Glycol mains to ramps, 2-1/2" (allowance, not drawn)	-400	lf	-\$14,000	-240	-\$14,400	
Snow melt manifolds and controls connections	-2	ea	-\$3,000	-24	-\$1,440	
Subtotal			-\$17,000	-264	-\$15,840	
Overhead 20%, misc 2%, tax 8.875% on material						-\$8,760
VE-4 total (rounded)						-\$41,600

Electric: 0 / +240 kW if electric by EC. Approvals: Owner decision; civil / landscape for drainage.

VE-5 Pool and spa: dedicated gas heaters in lieu of HX-1 / HX-2

Drawings: M5000, M6000 (HX-1 pool 500 MBH, HX-2 spa 245 MBH)

Delete pool and spa heat exchangers and their boiler-side piping. Dedicated condensing gas pool and spa heaters with venting (gas piping by plumber, excluded). Deduct only if heaters move to pool contractor.

Line item	Qty	Unit	Material	MH	Labor	Services
HX-1 pool heat exchanger	-1	ea	-\$6,000	-24	-\$1,440	
HX-2 spa heat exchanger	-1	ea	-\$4,000	-24	-\$1,440	
Boiler-side piping to HX-1 / HX-2, 2" (allowance)	-150	lf	-\$4,500	-75	-\$4,500	
Gas pool heater ~600 MBH input, condensing	1	ea	\$14,000	32	\$1,920	
Gas spa heater ~300 MBH input, condensing	1	ea	\$8,000	24	\$1,440	
Heater venting, sidewall / areaway	2	ea	\$6,000	32	\$1,920	
Subtotal			\$13,500	-35	-\$2,100	
Overhead 20%, misc 2%, tax 8.875% on material						\$3,700
VE-5 total (rounded)						\$15,100

Electric: 0. Approvals: Pool designer; NY All-Electric Buildings Act check for new gas appliances.

VE-6 Delete boiler plant and hot water / glycol distribution

Drawings: M2001-M2003, M2014-M2016, M5000, M6000

Possible only with VE-1 to VE-5. Deletes B-1/2/3, BP-1/2, PP-1/2, P-1 to P-4, HS-1, ET-1/2, AS-1, HX-3, chemical feed, glycol fill, boiler room piping, basement mains and risers, 1st floor hot water piping, insulation, boiler venting, combustion air, water treatment, hydronic balancing and start-up. GC also saves the fire-rated chimney enclosures and combustion air areaway.

Line item	Qty	Unit	Material	MH	Labor	Services
Boilers B-1, B-2, B-3 (1,999 MBH each), set and connect	-3	ea	-\$204,000	-240	-\$14,400	
Pumps BP-1/2, PP-1/2	-4	ea	-\$32,000	-64	-\$3,840	
Pumps P-1 to P-4 (glycol)	-4	ea	-\$18,000	-64	-\$3,840	
HS-1, ET-1/2, AS-1, chemical feeder, glycol fill package	-1	ls	-\$22,000	-60	-\$3,600	
HX-3 glycol heat exchanger	-1	ea	-\$9,000	-24	-\$1,440	
Boiler room piping 2-1/2" to 6" welded (allowance, not drawn)	-400	lf	-\$22,000	-480	-\$28,800	
Boiler room valves, trim, gauges (allowance)	-1	ls	-\$25,000	-120	-\$7,200	
Basement mains and risers, 2-1/2" avg (allowance, not drawn)	-1,200	lf	-\$42,000	-720	-\$43,200	
Glycol piping to garage / basement units, 1-1/2" (allowance)	-900	lf	-\$19,800	-360	-\$21,600	
1st floor HW piping 1/2" to 1" (measured, M2014-M2016)	-3,400	lf	-\$47,600	-1,020	-\$61,200	
1st floor HW mains 1-1/4" to 2-1/2" (measured, M2014-M2016)	-1,500	lf	-\$42,000	-675	-\$40,500	
Pipe insulation, hot water and glycol	-7,800	lf	-\$39,000	-624	-\$37,440	
Boiler venting, basement to roof (3 boilers)	-200	lf	-\$18,000	-100	-\$6,000	
Combustion air duct, louver, damper	-1	ls	-\$4,000	-40	-\$2,400	
Flush, fill, water treatment, pressure test	-1	ls	-\$6,000	-80	-\$4,800	
Boiler factory start-up	-1	ls				-\$6,000
Hydronic balancing (3rd party)	-1	ls				-\$18,000
Subtotal			-\$550,400	-4,671	-\$280,260	-\$24,000
Overhead 20%, misc 2%, tax 8.875% on material						-\$236,840
VE-6 total (rounded)						-\$1,091,500

Electric: -. Approvals: IMEG; Sunrise.

VE-9 Fire/smoke dampers at apartment duct branches

Drawings: M2020-M2043, M2100-M2110 (about 360 FSD tags)

DOAS supply branches to each apartment (6x4 to 8x6) are drawn with combination fire/smoke dampers. Branches under 100 sq in in 26 ga steel, no openings into the corridor, above the ceiling, with 12" steel sleeve: no damper required at corridor fire partitions (IBC 717.5.4 exception). Dampers stay at shafts and smoke barriers. EC and fire alarm credits for damper power and modules are additional.

Line item	Qty	Unit	Material	MH	Labor	Services
Combination fire/smoke dampers at apartment branches	-283	ea	-\$183,950	-849	-\$50,940	
Access doors at dampers	-283	ea	-\$16,980	-142	-\$8,490	
Steel sleeves at corridor wall penetrations	283	ea	\$11,320	283	\$16,980	
Subtotal			-\$189,610	-708	-\$42,450	
Overhead 20%, misc 2%, tax 8.875% on material						-\$67,840
VE-9 total (rounded)						-\$299,900

Electric: -. Approvals: IMEG / HKS code analysis (AL Group I-1, smoke barriers), building department.

VE-14 Apartment bath exhaust ducted to DOAS exhaust

Drawings: M2020-M2043, M2100-M2110, M6001 (TX-A)

Each apartment bath has its own exhaust fan TX-A (75 CFM) ducted out the exterior wall. Duct bath exhaust to the DOAS exhaust trunk in the corridor instead (energy wheel recovers it). Deletes 283 fans, wall caps and EC circuits. Corridor return trunk is reused as the exhaust trunk.

Line item	Qty	Unit	Material	MH	Labor	Services
TX-A bath exhaust fans	-283	ea	-\$50,940	-708	-\$42,450	
Wall caps and exhaust duct to exterior wall	-283	ea	-\$29,715	-424	-\$25,470	
Exhaust branch to corridor trunk with grille (no damper, VE-9)	283	ea	\$33,960	849	\$50,940	
Corridor exhaust trunk upsizing (allowance)	1	ls	\$30,000	400	\$24,000	
Subtotal			-\$16,695	117	\$7,020	
Overhead 20%, misc 2%, tax 8.875% on material						-\$3,625
VE-14 total (rounded)						-\$13,300

Electric: -11 kW. Approvals: IMEG; DOAS exhaust airflow check.

Open items (not priced)

VE-11 No BMS: standalone equipment controls only (Sunrise Northport precedent). Deduct = BMS amount carried in quote 3112 less standalone controls. Needs the BMS contractor's breakdown. Supersedes VE-7.

VE-12 Common areas: delete 47 VAV boxes; single-zone RTUs for large spaces, ductless / VRF for small rooms. Replaces VE-1 if taken. Priced after IMEG issues a zoning concept.

VE-10 Shorten DOAS corridor return. Included in VE-14 (return trunk reused as exhaust).